Unit 4 · Outcome 2 · Energy Sources

Fossil fuel use & peak oil

KK04 VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.

Fossil fuel use & peak oil

U4·O2 · KK04 Unit 4 · Energy Sources · Year 12 Fossil fuel use & peak oil "Changes in the rate of fossil fuel use over time and the concept of peak oil." 3D view unavailable — the explanation below stands on its own. A Hubbert curve: extraction rate rises, peaks when about half the resource is gone, then declines.

A1Three fossil-fuel eras

A1 Three fossil-fuel eras For almost all of human history, our energy came from muscle, wood, wind and water. Fossil fuels changed that — but they did not all arrive at once. They scaled up in sequence, each unlocked by a new technology. ① Coal — the steam age From the Industrial Revolution (~1760s onward), coal powered steam engines, iron-making and railways. It was the first fossil fuel to scale, and it dominated global energy into the early 20th century. ② Oil — the mobile age After the first commercial well (1859) and the internal combustion engine

A2What pushes the rate up — and back down

A2 What pushes the rate up — and back down If a question asks you to explain changes in the rate, you need causes, not just a description of the line. Examiners want the drivers on both sides. ▲ Drives the rate up Population growth — more people, more energy demanded Industrialisation & economic growth — factories, construction, freight Rising living standards — appliances, cars, air-con, travel Transport & electrification spreading worldwide Cheap, energy-dense, easy to store — fossil fuels were simply convenient ▼ Bends the rate down Energy efficiency

B1Peak oil: Hubbert and the curve

B1 Peak oil: Hubbert and the curve Now the second half. "Peak oil" is a precise idea — and the single most-tested thing about it is what it does not mean. Definition — learn this wording Peak oil The point in time at which the maximum rate of oil extraction (production) is reached, after which the rate of production enters a terminal, irreversible decline . In 1956 , geoscientist M. King Hubbert reasoned that because oil is a finite, non-renewable resource, its production over time must trace a roughly symmetrical bell-shaped curve — now called a Hubbert

B2Why the global peak keeps moving

B2 Why the global peak keeps moving Hubbert nailed the US peak, but global "peak oil" has been predicted — and missed — many times. Knowing why turns a 2-mark recall into a 4-mark explanation. Factor How it pushes the peak later (or reshapes the curve) New discoveries Fields once thought not to exist (deepwater, Brazil's pre-salt) added recoverable oil, lifting and stretching the curve. Better extraction technology Horizontal drilling and hydraulic fracturing unlocked US shale / tight oil from ~2010, transforming the supply picture and delaying the suppo

Factor	How it pushes the peak later (or reshapes the curve)
New discoveries	Fields once thought not to exist (deepwater, Brazil's pre-salt) added recoverable oil, lifting and stretching the curve.
Better extraction technology	Horizontal drilling and hydraulic fracturing unlocked US shale / tight oil from ~2010, transforming the supply picture and delaying the supposed peak.
Unconventional sources	Oil sands, tight oil, coal seam gas and oil shale count as oil-equivalent supply once technology and prices allow.
Price	High prices make difficult, low-grade fields economic, so reserves that were "uneconomic" suddenly count.
Demand	Efficiency, EVs and renewables can lower demand — which, as we'll see, can bring the peak forward from the other side.

B3Peak supply vs peak demand

B3 Peak supply vs peak demand The original peak-oil idea was a supply story: production peaks because the resource is finite. But the conversation has flipped. Many analysts now argue the peak may arrive because demand falls away first — driven by electric vehicles, efficiency, renewables and climate policy — rather than because we can't pump any more. ▶ Analogy — why did the party end? Did the music stop because the band ran out of songs (peak supply) or because the crowd went home (peak demand)? Both empty the dancefloor — but for opposite reasons. M

★Bass Strait: a Hubbert curve on Victoria's doorstep

★ Bass Strait: a Hubbert curve on Victoria's doorstep You don't need to look overseas for peak oil. Off the Gippsland coast — the view from our case-study town, Tarra Cove — the real Bass Strait fields tell the whole story in miniature. The rise and the peak The Gippsland Basin (Bass Strait), run by Esso/ExxonMobil with Woodside, has produced oil and gas for more than 50 years . Its oil output rose, peaked and is now in mature decline — about half of all the wells no longer produce, and decommissioning of the spent platforms begins in 2027 . A textbook H

01Earlier or later?

01 Earlier or later? Drag each factor into the bin showing what it does to a global peak-oil date . This is the exact reasoning a 4-mark "explain why predictions changed" question wants. Discovery of new deepwater oil fields Hydraulic fracturing unlocks shale oil Rapid uptake of electric vehicles Oil price rises, making hard fields economic Strong renewable + efficiency policy Tapping oil sands & tight oil Falling demand for petrol cars Pushes the peak LATER Brings the peak EARLIER Check

02Decode the command word

◈ COMMAND WORDS

02 Decode the command word The verb tells you what shape your answer must take. Tap one to see what KK04 expects. Describe Explain Distinguish Outline Evaluate Justify

03Six exam traps

⚠ EXAM TRAPS

03 Six exam traps Every one of these has cost real students real marks. Read them as a pre-flight checklist. ✕ Trap 1 — "Peak oil means we're running out" The single biggest error. Peak oil is the peak of the extraction rate ; about half the resource remains at the peak. Say rate , not "amount left". ✕ Trap 2 — Describing the curve instead of explaining it "Explain" needs causes . Don't just say "use went up then levelled off" — name drivers (industrialisation, transport, then efficiency, renewables, policy). ✕ Trap 3 — Treating "fossil fuels" as one thi

04P·E·L trainer

✎ P·E·L WRITING

04 P·E·L trainer Build a high-band answer in three moves — Point , Evidence , Link . Type into each box; the trainer flags the key ideas examiners look for. Practice prompt 4 marks Explain the concept of peak oil, and explain one reason why predictions of a global oil peak have repeatedly been pushed back. (4 marks) P · E · L · Reveal a model P·E·L answer Point: Peak oil is the point at which the global rate of oil extraction reaches its maximum and then enters terminal decline — it refers to the production rate, not to oil running out. Evidence: At the

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

05Same question, two answers

✗ WEAK ANSWER

✕ Weak — 1 / 3 "Peak oil is when oil starts to run out and there isn't much left. It doesn't mean we run out straight away because there's still some oil in other countries we can use."

✓ STRONG ANSWER & MARKING

✓ Strong — 3 / Why it loses marks: the definition is wrong — it equates peak oil with running out, the exact trap. No mention of extraction rate , no "half remains" idea, and the second sentence is vague geography rather than the real reason.

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

06Practice exam questions

★ PRACTICE & SELF-MARK

06 Practice exam questions Five VCAA-style questions · 20 marks total Write your answer, reveal the marking guide, then self-mark. Your score saves automatically and feeds the dock. Q1 Define the term peak oil . 2 marks Show marking guide Marking guide & model answer Peak oil is the point in time at which the maximum rate of oil extraction (production) is reached (1) , after which the rate of production enters a permanent / terminal decline (1) . Marking points 2/2 Identifies it as the peak of the rate of extraction, and notes the subsequent decline. 1/2

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Peak oil is when oil starts to run out and there isn't much left. It doesn't mean we run out straight away because there's still some oil in other countries we can use." Why it loses marks: the definition is wrong — it equates peak oil with running out, the exact trap. No mention of extraction rate , no "half remains" idea, and the second sentence is vague geography rather than the real reason.

✓ STRONG / MODEL ANSWER

Peak oil is the point at which the maximum rate of oil extraction is reached, after which production permanently declines. It does not mean oil is running out, because at the peak roughly half the recoverable resource still remains underground. What declines is the rate of production, as the cheap, easy oil is used first and the remainder is harder and costlier to extract." Why it scores: precise definition built on rate ; directly tackles the misconception; uses the half-remains fact; explains the mechanism (easy oil first). Three clean, separable marking points.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.